

7 An algorithm has been written in pseudocode to:

- input a whole number between 1 and 255 inclusive
- convert the input to an 8-bit binary number
- store each bit of the binary number in an array.

For example, if the denary number 127 was input, the binary number 01111111 would be stored in the array.

```
01 DECLARE BinaryArray : ARRAY[1:8] OF INTEGER
02 DECLARE DenaryNumber : CHAR
03 OUTPUT "Please enter a whole number between 1 and 255 inclusive "
04 INPUT DenaryNumber
05 FOR Count ← 7 TO 1 STEP -1
06     BinaryArray[Count] ← MOD(DenaryNumber,3)
07     DenaryNumber ← DIV(DenaryNumber,2)
08 NEXT Count
```

(a) Identify the line numbers of the **three** errors in the pseudocode and suggest a correction for each error.

Error 1 line number

Correction

Error 2 line number

Correction

Error 3 line number

Correction

[3]

